Math 1220
Unit 2: *Polynomial, Rational, Inverse, Exponential & Logarithmic Functions*
Sections 7.2, 8.1-8.2, 9.1-9.7

7.2 Polynomial Functions

Before considering polynomials in general, we will first look at the Power Functions or Monomial Polynomials.

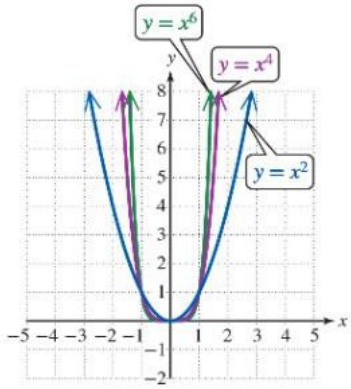
Power Functions

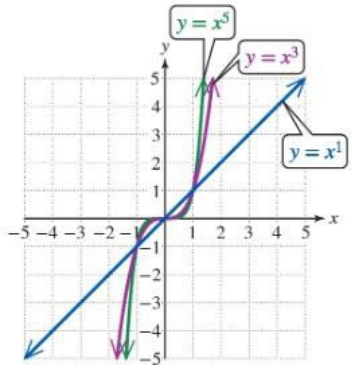
Monomial Polynomials, $p(x) = x^n$

Properties of $f(x) = x^n$

- If n is even, then the graph is similar to $y = x^2$
- If n is odd, then the graph is similar to $y = x^3$

For any n, as n increases in magnitude, the graph is steeper for $x < 1$ and $x > 1$, but between -1 and 1 the graph flattens out.





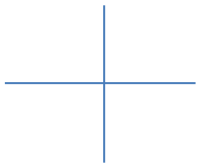
Transformations of Power Functions, $p(x) = a(x-h)^n + k$

Example 1 Graph each power function by applying transformations (just reflections and shifting – no stretching!) of the basic curve. Determine the intercepts and state the domain and range.

(a) $p(x) = \frac{1}{4}(x-2)^5 - 1$

Basic shape (dotted)

Reflection: (solid)



Shifts:

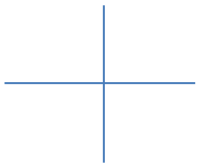
"Vertex" or "Center point"

***y*-int:** ***x*-int:**

(b) $f(x) = -\frac{1}{2}(x+1)^4 + 8$

Basic shape (dotted)

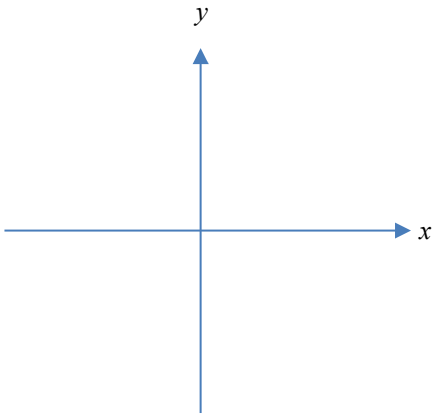
Reflection: (solid)

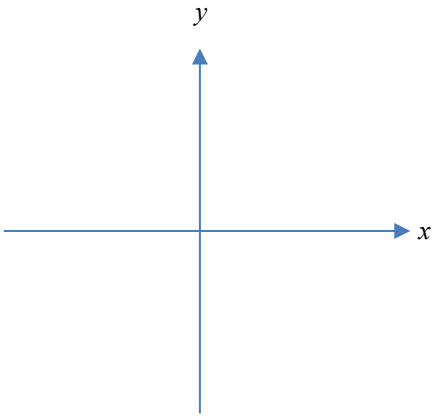


Shifts:

"Vertex" or "Center point"

***y*-int:** ***x*-int:**

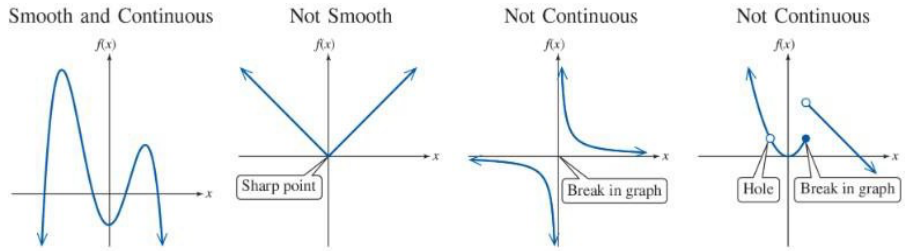




Now we will look at polynomial functions in general.

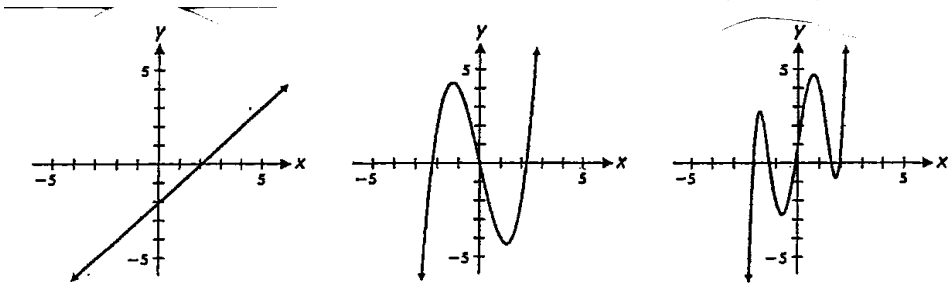
Polynomial Functions are smooth and continuous, meaning no sharp points or breaks in the graph. Only the first graph shown is a polynomial.

The **domain** for a polynomial is all real numbers.

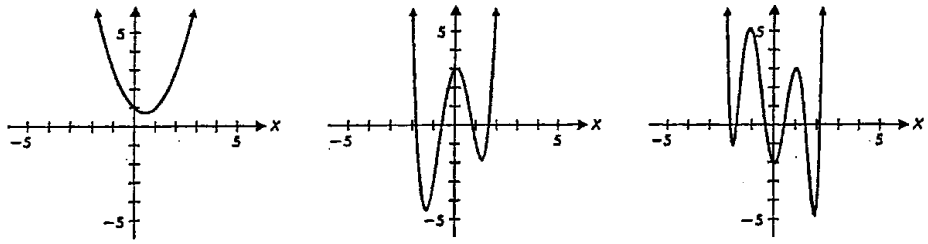


<u>The Polynomial Pattern</u> n^{th} **degree polynomial**: $p(x) = a\ x^n + a\ \ x^{n-1} + \ldots + a\ x + a$

$n \qquad n-1 \qquad 1 \qquad 0$



(a) $f(x) = x - 2$ (b) $f(x) = x^3 - 5x$ (c) $f(x) = x^5 - 6x^3 + 8x + 1$



(d) $f(x) = x^2 - x + 1$ (e) $f(x) = 2x^4 - 7x^2 + x + 3$ (f) $f(x) = x^6 - 7x^4 + 12x^2 - x - 2$

Turning Point (TP): A point on a continuous graph that separates an increasing portion from a decreasing portion.

1) **Degree of polynomial *vs* # of TPs and *x*-ints:**
An *n*th degree polynomial has *at MOST* (1) *n*-1 turning points and (2) *n* *x*-intercepts.

2) End behavior (EB):
<u>odd degree</u> – arrows go opposite direction;
<u>even degree</u>, arrows go same direction.

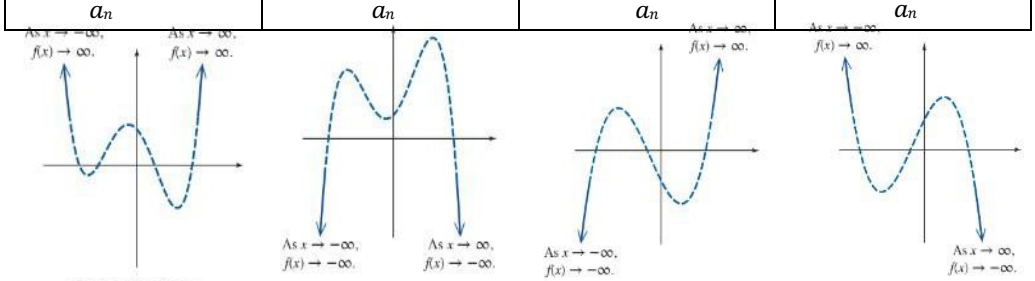
3) Note: the lead coefficient for all is positive. What would happen if it was negative?

Graph Properties of an *nth* degree Polynomial Function *P*

1. *P* is continuous for all real numbers and the curve is smooth.
2. *P* has at most *n* *x*-intercepts and at most *n* – 1 turning points.
3. **End behavior**: If *n* is even, both ends of the graph point in the same direction.
   If *n* is odd, the ends of the graph point in opposite directions.
4. *P* has a local minimum or maximum at each turning point.

Leading Term Test: The general shape of $f(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0$ depends on the leading term, $a_n x^n$.

***n* even**		***n* odd**	
a_n	a_n	a_n	a_n



End behavior: up left/up right

End behavior: down left/down right

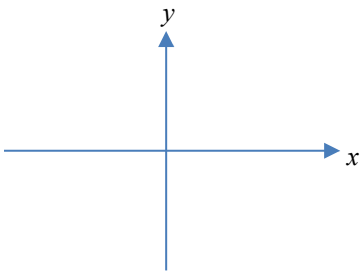
End behavior: down left/up right

End behavior: up left/down right

As $x \to \infty$ or as $x \to -\infty$, the graph of *f* will eventually have no more turns and will become forever increasing or forever decreasing. Thus, the graph of *f* far to the left or right will follow the general behavior of $y = a_n x^n$

Example 2A Graph a polynomial with its maximum number of turning points that satisfy the given requirements.

(a) -5th degree, exactly 3 x-intercepts

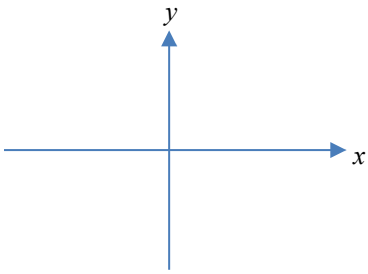


End behavior:

As $x \to -\infty, f(x) \to$ _____

As $x \to +\infty, f(x) \to$ _____

(b) +6th degree, only 1 x-intercept

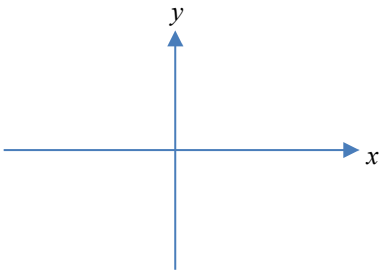


End behavior:

As $x \to -\infty, f(x) \to$ _____

As $x \to +\infty, f(x) \to$ _____

(c) -4th degree, no x-intercepts



End behavior:

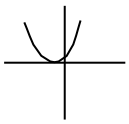
As $x \to -\infty, f(x) \to$ _____

As $x \to +\infty, f(x) \to$ _____

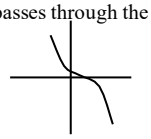
Your Turn 1: Follow instructions in **Example 2A:** (a) -6th degree, exactly 3 x-intercepts (b) +5th degree, (3, 0) is on the graph

Behavior of Curve at a Zero

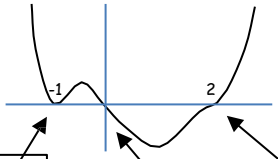
If the zero has **even** multiplicity, then the curve touches the zero and turns back.



If the zero has **odd** multiplicity, then the curve passes through the zero.



$$P(x) = x(x-2)^3(x+1)^2$$



the zero of m_2 causes the curve to "touch & turn."

zero of m_1 passes through w/o changing shape

zero of m_3 causes a flattening-like "bobble"

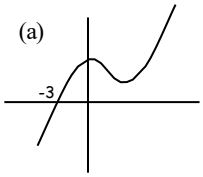
Multiplicity of a Zero
For this polynomial, we say that

0 is a zero of multiplicity 1 (m_1)
2 is a zero of multiplicity 3 (m_3)
-1 is a zero of multiplicity 2 (m_2)

An nth degree polynomial (degree > 0) has exactly n zeroes (counting multiplicity).

Example 2B For each polynomial curve, determine the sign and smallest possible degree of the leading term, and the number of real and imaginary zeros. For each real zero, state its multiplicity.

(a)



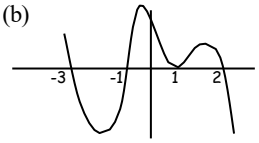
Degree/Sign: _____

real zeros: _____

imag zeros: _____

Real Zero Mult.

(b)



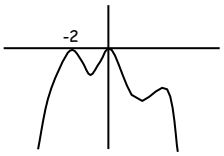
Degree/Sign: _____

real zeros: _____

imag zeros: _____

Real Zero Mult.

(c)



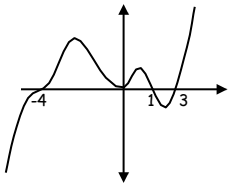
Degree/Sign: _____

real zeros: _____

imag zeros: _____

Real Zero Mult.

(d)



Degree/Sign: _____

real zeros: _____

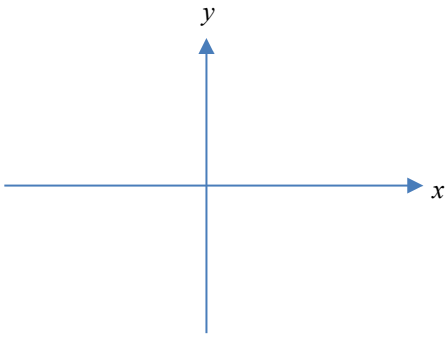
imag zeros: _____

Real Zero Mult.

Write a feasible equation for the function shown in (b):

Example 3 Graph. 1) Determine the leading term and end behavior.
2) Mark the real zeros and y-intercept as numbers on the axes.
3) Test points between the zeros to see whether the curve is above or below the x-axis.

(a) $h(x) = -x(x+2)^3(x-3)^2$



Use the graph to solve
$-x(x+2)^3(x-3)^2$

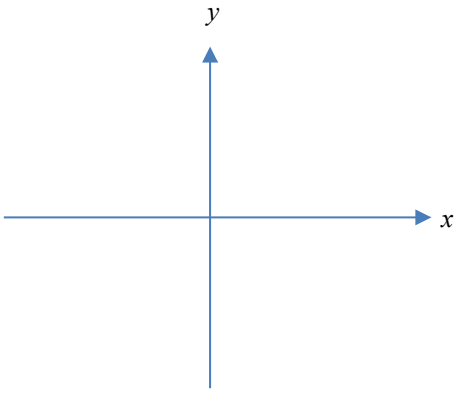
Lead term End Behavior

x	-3	-2	-1	0	1	3	4
$p(x)$		0		0		0	

(b) $m(x) = x^4 - 2x^2 + 1$

Factor $m(x)$:

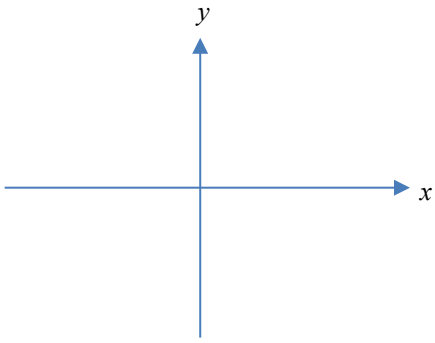
Lead term End Behavior



(c) $g(x) = -x^3 + 4x^2 - 4x$

Factor $g(x)$:

Lead term End Behavior



Your Turn 2: Follow instructions from **Ex3** above:
(a) $p(x) = -2x(x+1)^3(x-2)^2$, (b) $k(x) = -6x^3 + 26x^2 - 28x$

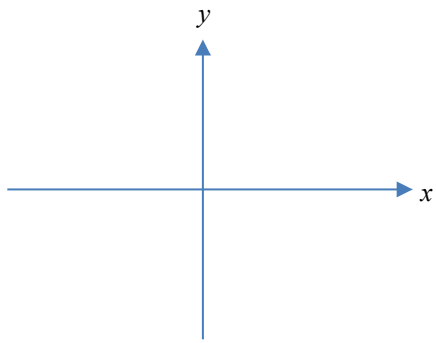
MP2 7.2 #25, 27, 35, 37, 59, 60, 63, 71, 72

Power functions *vs* the Others

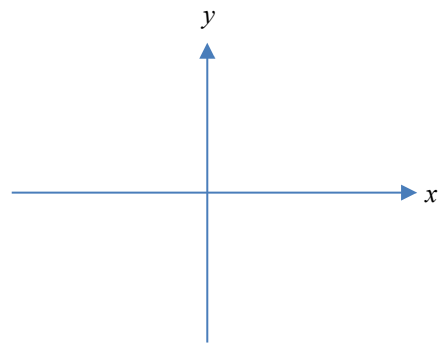
$p(x) = a(x-h)^n + k$ *vs* $p(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0$ or $p(x) = a(bx+c)^m(dx+e)^p$ ...

Example 4 Graph. Mark all intercepts and important points.

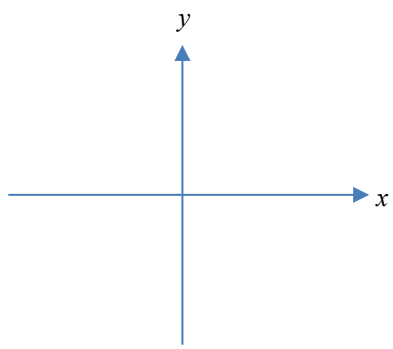
(a) $p(x) = \frac{1}{4}(x+1)^5 - 8$



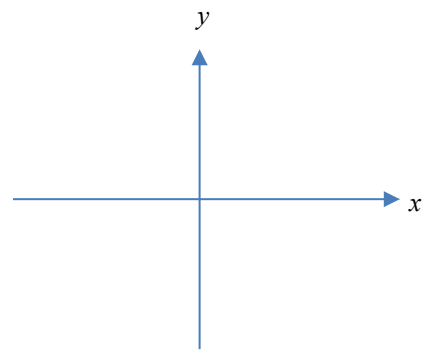
(b) $f(x) = \frac{1}{4}(x+1)^2(x-2)^2(x-3)$



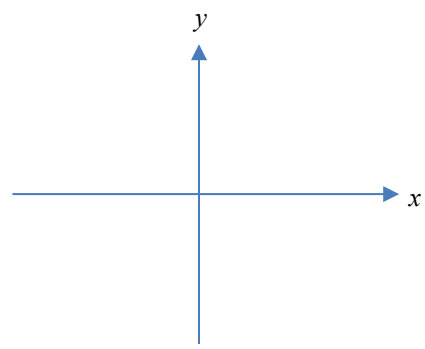
(c) $t(x) = \frac{1}{2}(x+1)^4 - 8$



(d) $g(x) = 2x^4 + 5x^2 - 3$



(e) $f(x) = 2x^4 + 5x^3 - 3x^2$

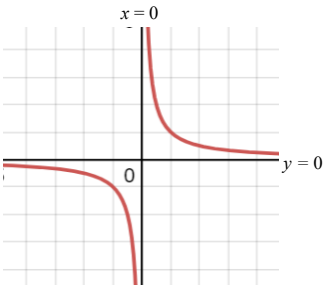


MP3 Find all real zeros and graph. (a) $p(x) = \frac{1}{4}(x+1)^3 - 2$ (b) $h(x) = x^4 - 1\ x^2 + 9$

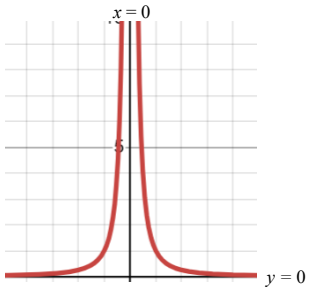
8.1-8.2 Rational Functions

Basic Rational Curves

$y = \frac{1}{x}$



$y = \frac{1}{x^2}$



	HA: As x →±∞, y →b. So y = b is a HA							
	EB: As x →∞, y →0				**EB:** As x →-∞, y →0			
x	1	5	10	100	1	5	10	100
$1/x^2$	1	1/5	1/10	1/100	-1	-1/5	-1/10	-1/100
	1	1/25	1/100	1/10000	1	1/25	1/100	1/10000

	VA: As x →a, y →±∞. So x = a is a VA					
	As x →0 from the right and left sides					
x	1/10	1/100	As x → 0^+	-1/10	-1/100	As x → 0^-
1/x	10	100	y → ∞	-10	-100	y → -∞
$1/x^2$	100	10000	y → ∞	100	10000	y → ∞

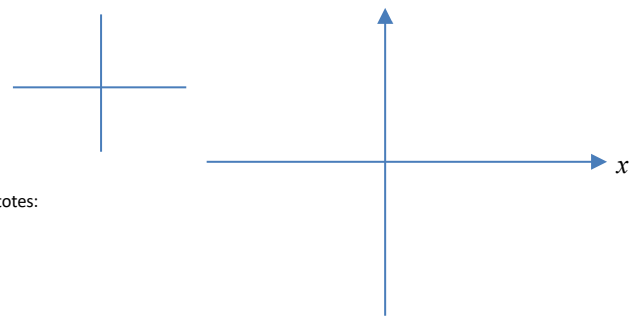
Example 5 Graph. (1) Determine any reflections. (2) Determine the shifts which lead to the asymptotes. (3) Find intercepts.

(a) $f(x) = \frac{4}{2-x} + 1$

Basic shape (dotted)

Reflection: (solid)

Shifts/Asymptotes:



Vertical Asymptote or a Deleted Point?

A vertical asymptote occurs at any x value where the denominator of a REDUCED function is zero. In other words, a rational function has a vertical asymptote where the denominator is 0 *unless* it "cancels out."

If it cancels out, then that x value becomes the location of a DELETED POINT, DP (hole in the graph), and NOT a VA.

Example 6 Determine all the "x-stuff": domain, x-intercepts and vertical asymptotes.

(a) $y = \frac{(x-2)(x+1)}{(x-3)(x+4)}$ (b) $y = \frac{3}{x^2-9}$ (c) $y = \frac{x(x-4)}{x^2+x}$ (d) $y = \frac{x^2}{x^2+x+1}$

D: _____
VA: _____
x-int: _____
DP: _____

D: _____
VA: _____
x-int: _____
DP: _____

D: _____
VA: _____
x-int: _____
DP: _____

D: _____
VA: _____
x-int: _____
DP: _____

Horizontal Asymptotes:

$$f(x) = \frac{a_n x^n + \ldots}{b_m x^m + ..}$$

(1) If deg(num) < deg(den), **then** $y = 0$ **is the HA**.

(2) If deg(num) = deg(den), **then** $y = \frac{a_n}{b_m}$ **is the HA.**

(3) If deg(num) > deg(den), **then there is no HA**.

(4) If deg(num) is exactly one more than deg(den), **then there is a slant asymptote, SA.**

When it comes to horizontal asymptotes, it's ALL about the *power* of the top versus the bottom. So, to find the HA, you can trim a function down to its largest degree terms from the top and bottom of the fraction. The function will behave like $y = \frac{a_n x^n}{b_m x^m}$ for large values of $|x|$

(1) $f(x) = \frac{4x^2-2x}{5x^3+1}$

(2) $f(x) = \frac{4x^2-2x}{5x^2+1}$

(3) $f(x) = \frac{4x^2-2x}{5x+1}$

Example 7 For each function, determine the *y*-intercept and any horizontal or slant asymptotes (SA) that exist.

(a) $f(x) = \frac{1-x^2}{2x^2+3x+1}$

(b) $g(x) = \frac{x+2}{4x-3x^2}$

(c) $h(x) = \frac{x^2+2x-5}{x-1}$

y-int:

HA:

y-int: _____

HA: _____

SA: _____

y-int: _____

HA: _____

SA: _____

y-int: _____

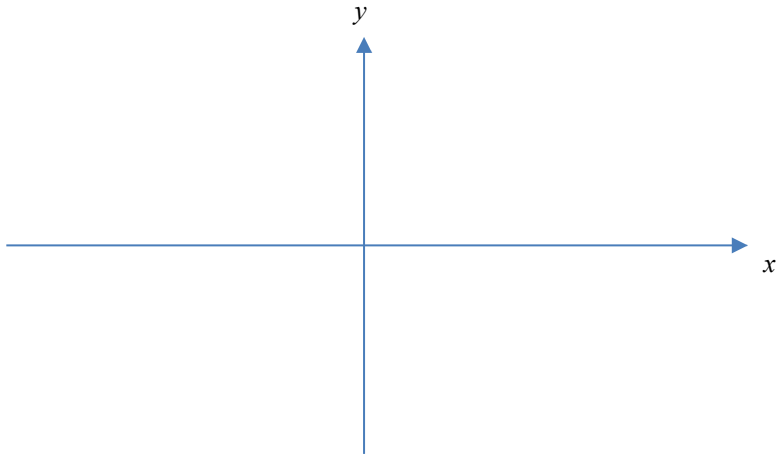
HA: _____

SA: _____

Graphing: 1. FACTOR the function. Then find all intercepts, asymptotes and deleted points that exist. Put on graph.
2. Determine whether the function is above or below the x-axis over critical intervals determined by x-intercepts and VAs.

Example 8 Graph each function as outlined above. Justify why the curve is above or below the x-axis on each critical interval.

(a) $f(x) = \frac{3x-6}{2x+7}$



x-int: _____

VA: _____

y-int: _____

HA: _____

DP: _____

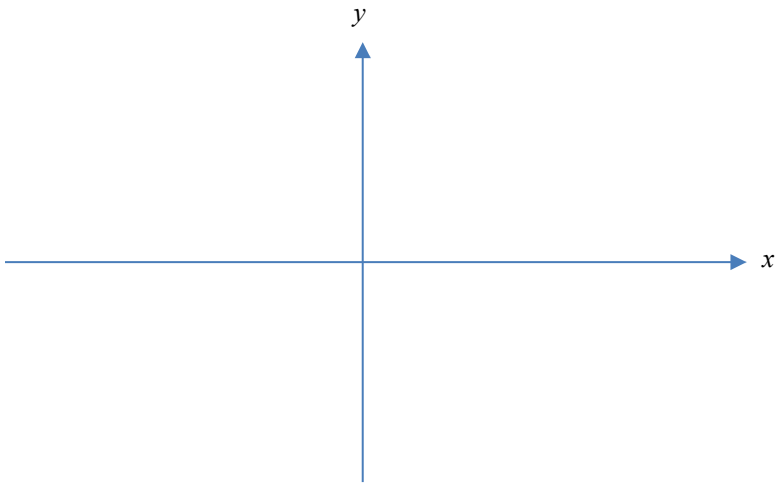
Solve $\frac{3x-6}{2x+7} \geq$

x-intervals:

Above or Below:

Justification:

(b) $g(x) = \frac{2x^2-x-3}{x^2-2x-3}$



x-int: _____

VA: _____

y-int: _____

HA: _____

DP: _____

x-intervals:

Above or Below:

Justification:

Example 9 Graph each function. Completely FACTOR the function 1st and justify why you have the function above/below the x-axis over each critical interval.

(a) $k(x) = \frac{x^2+4x+4}{x^2-2x-3}$

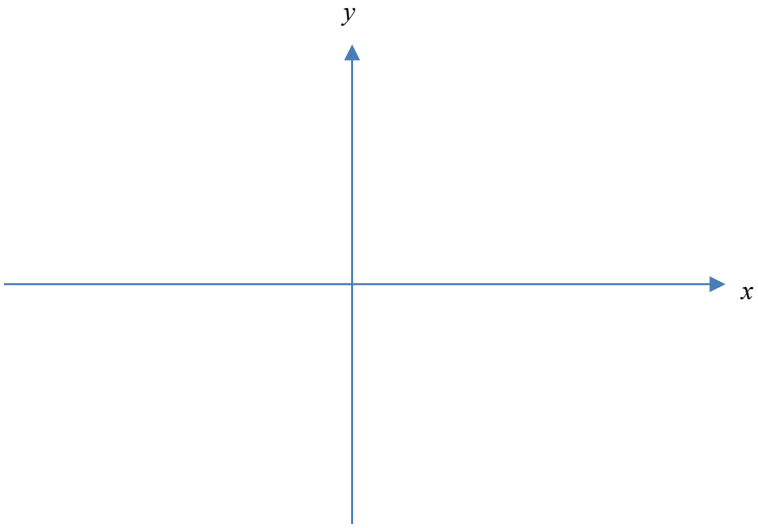
x-int: _____

VA: _____

y-int: _____

HA: _____

DP: _____



x-intervals:

Above or Below:

Justification:

(b) $h(x) = \frac{x^2+x-6}{x^3+3x^2}$

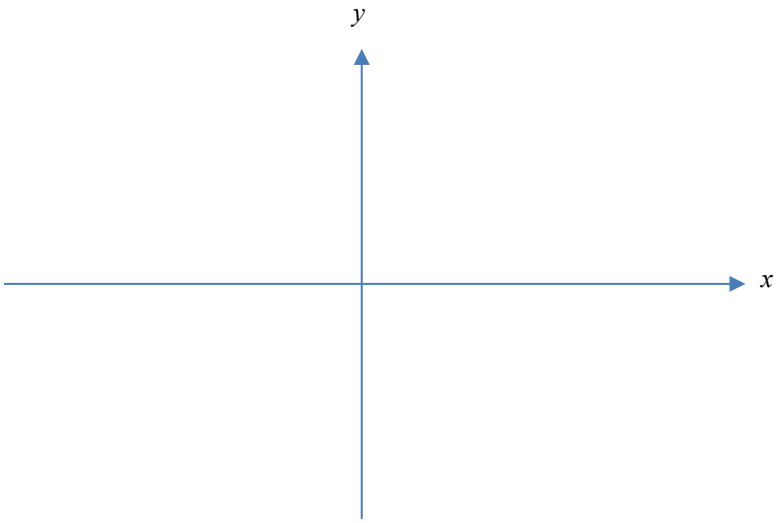
x-int: _____

VA: _____

y-int: _____

HA: _____

DP: _____



x-intervals:

Above or Below:

Justification:

MP5A 8.2 #21, 22

MP5B For each, find all intercepts, asymptotes & deleted points that exist, and graph. You must (1) completely FACTOR the function 1st and (2) justify why you have the function above/below the x-axis over each interval.

(a) $f(x) = \frac{2x+6}{3x-5}$ (b) $g(x) = \frac{x^2-2x+1}{x^2-4x+4}$ (c) $k(x) = \frac{x(x-1)^2}{x^3-3x^2-4x}$ (d) $r(x) = \frac{2x}{x^2+2x+1}$

9.2 Exponential Functions

An *exponential function* is a function in which the exponent is the variable and the base is a constant. This is very different from an algebraic function where the variable is the base and the exponent is a constant. An example follows.

Algebraic function: $y = x^2$ Exponential function: $y = 2^x$

The word "exponential" should convey a very quick or steep increase (or decrease). Relatively speaking, exponential functions increase or decrease very fast compared to other functions. It might be a good idea to re-familiarize yourself with the laws of exponents.

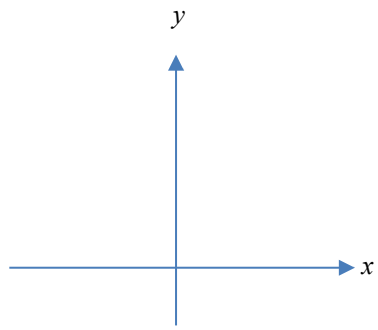
Exponential Function: $f(x) = b^x$ where b , $b \neq 1$. The domain of f is all real numbers.

Why the restrictions on the base, b?

The Graph of $y = b^x$

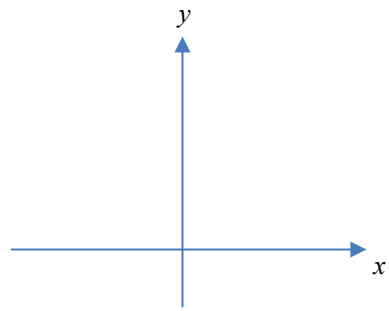
I. $b > 1$

$y = 2^x$



II. $0 < b < 1$

$y = (\frac{1}{2})^x$



$f(x) = b^x$

D: _____

R: _____

HA: _____

y-int (CP): _____

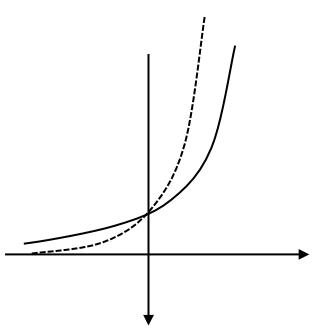
$As\ x \to +\infty, 2^x \to$ _____

$As\ x \to -\infty, 2^x \to$ _____ . $\therefore$ _____

$As\ x \to +\infty, (\frac{1}{2})^x \to$ _____ . $\therefore$ _____

$As\ x \to -\infty, (\frac{1}{2})^x \to$ _____.

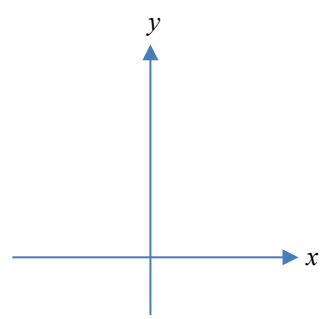
Example (a) Steepness: $y = 2^x\ vs\ y = 3^x$ (dotted)



x	2^x	3^x
0	1	1
1	2	3
2	4	8
-1	1/2	1/3
-2	1/4	1/9

(b) Base e $f(x) = e^x$

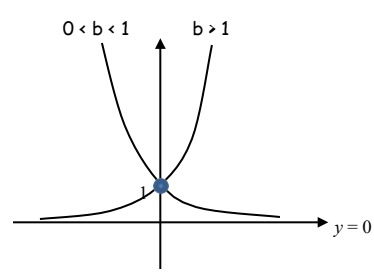
$e \cong 2.72$



x	f(x)
0	1
1	e
2	e^2
-1	$\frac{1}{e}$
-2	$1/e^2$

Basic Properties of the Graph of $f(x) = b^x$

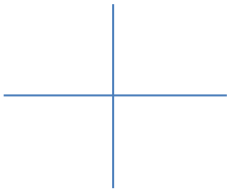
1. All pass through the point (0, 1)
2. All are continuous
3. HA: y = 0
4. If $b > 1$, then f is increasing over all x.
5. If $0 < b < 1$, then f is decreasing over all x.
6. The function f is one-to-one.ne



Transformations of the Basic Curve

Example 14 On the 1st small axes, draw the basic curve with a dotted line, and then the reflected curve with a solid line. Then determine the shifts and graph that on the 2nd larger axes. Determine the asymptote, range and y-intercept, and mark the new location of the critical point (0, 1).

(a) $f(x) = e^{2-x} - 1$

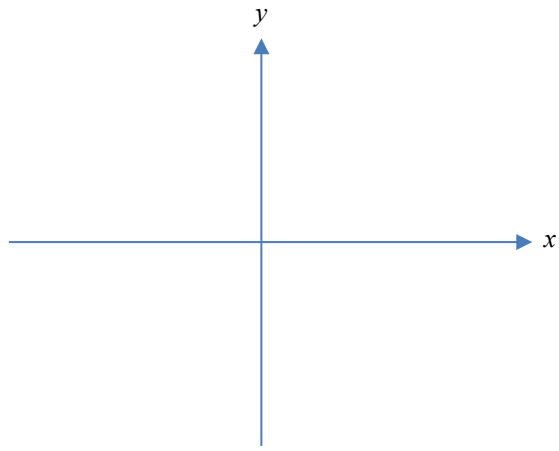


1) Basic curve: growth or decay?

2) Reflection?

Shifts:

Draw basic curve with dotted line, reflected curve with solid line.



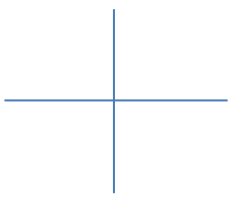
HA:

R:

y-int:

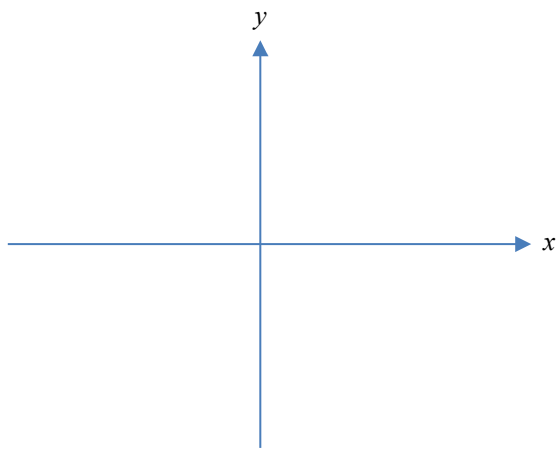
CP:

(b) $g(x) = 2 - (\frac{1}{3})^{x+1}$



Shifts:

Draw basic curve with dotted line, reflected curve with solid line.



HA:

R:

y-int:

CP:

1-1 Property for Exponents: $If\ b^M = b^N, then\ M = N$

Example 15 (a) Solve $4^{x-3} = 8^{2x+1}$

(b) Find the x-intercept of $f(x) = 3^{2-x} - 27$

9.3-9.4 Logarithmic Functions and Properties

The function $f(x) = b^x$ is a one-to-one function, so we know it has an inverse. So, let's try to find it.

Solving for the inverse of $f(x) = b^x$

$y = b^x$ Replace *f(x)* with *y*.

$x = b^y$ Swap *x* and *y* in the equation.

To find an inverse, we exchange *x* and *y* and then solve for *y*.
BUT...How do we solve for *y*?
We create the term ***logarithm*** to mean ***exponent***.

What is a Log? A **LOG** is an **EXPONENT!**

$x = b^y$

We say "*y* is the exponent on *b* that produces *x*."
Now we say "*y* is the *logarithm* on *b* that produces *x*."
So $y = \text{logarithm}_b x$
Abbreviated to $y = \log_b x$
which reads "*y* equals the *log* base *b* of *x*."

How about inverse of e^x?

Logarithmic Function: $f(x) = \log_b x$ *where* b *and* $b \neq 1$. The domain of *f* is all $x > 0$. The range of *f* is all real numbers.

Graphing $f(x) = \log_b x$

Since $y = \log_b x$ is the inverse function of $y = b^x$, their graphs are reflections about the line $y = x$.

The Graph of $y = \log_b x$

$f(x) = \log_b x$

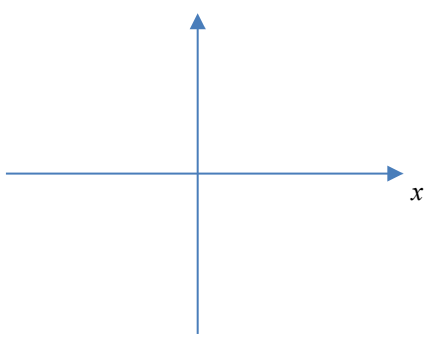
growth curve: $b > 1$ for example, $\boldsymbol{b} = \mathbf{2}$:

$f(x) = \log_2 x$

1st graph

$y = 2^x$

(dotted)



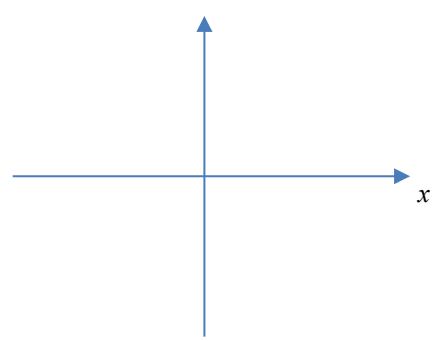
decay curve: $b < 1$ for example, $\boldsymbol{b} = ½$:

$f(x) = \log_{\frac{1}{2}} x$

1st graph

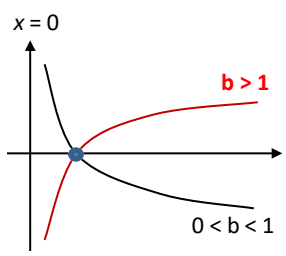
$y = (\frac{1}{2})^x$

(dotted)



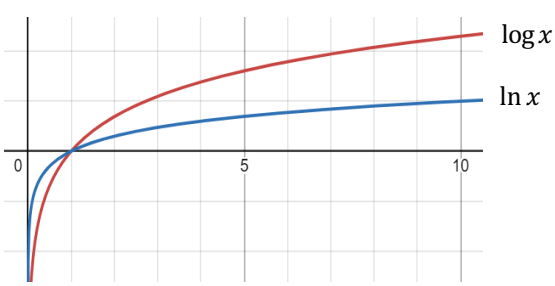
Basic Properties of the Graph of $f(x) = \log_b x$

1. Domain: $x > 0$
2. All pass through the point (1, 0)
3. VA: x = 0



	$y = b^x$	$y = \log_b x$
Domain		
Range		
VA		
HA		
CP		

Common logarithm: $\log x = \log_{10} x$
Natural logarithm: $\ln x = \log_e x$

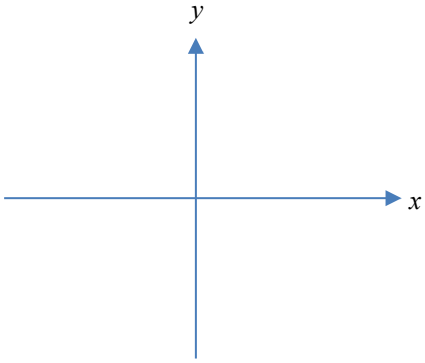


Example 16 Write the domain of $g(x) = \ln(3x + 2)$ in interval notation and determine the equation of the asymptote.

Example 17 Determine all intercepts. Give 2 other points other than intercepts/CP that can be evaluated without a calculator.

(a) $f(x) = \log_4(x+1) - 2$

Shifts:

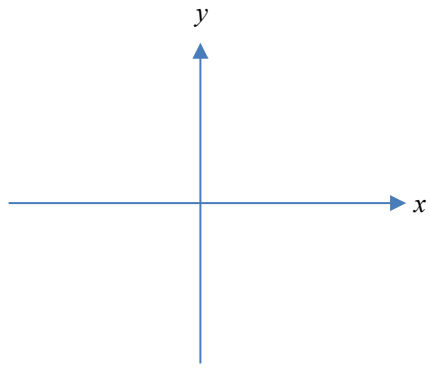


x-int:

y-int:

(b) $g(x) = 1 + \log_{\frac{1}{3}}(x+3)$

Shifts:



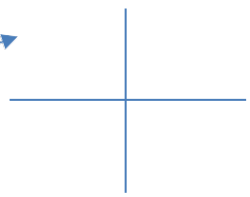
x-int:

y-int:

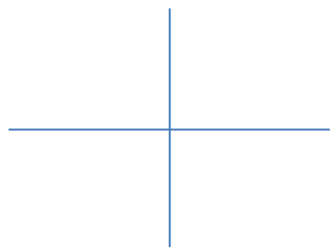
Example 18 Graph $g(x) = 1 - \ln(x+2)$. Determine the asymptote, domain and new location of the critical point (1, 0).

1) Basic curve: growth or decay?

2) Reflection?



Shifts:



VA:

D:

CP:

<u>Converting between Exponential & Log form</u>: $x = b^y$ is equivalent to $y = og_b x$

*There are NO coefficients on the exponential or log expression. You can NOT convert form if there is a coefficient!

Example 19 Convert from exponential form to logarithmic form or vice versa.

(a) $10^{-1} = \frac{1}{10}$ (b) $2^3 = 8$ (c) $e^x = 6$ (d) $\log_2 8 = 3$ (e) $\log 100 = 2$ (f) $2\ln x = 4$

Example 20 Find the x-intercept of each function.

(a) $f(x) = \frac{1}{2}\log_2(3-x) + 2$

(b) $g(x) = 2e^{x+1} - 3$

Example 21 Find the inverse of $g(x) = 2 + \ln(x - 1)$.

1. Put in a *y* for *g(x)*:

2. Swap *x* and *y*:

3. Solve for *y*: - Isolate *y*-term:

- Convert form:

- Isolate *y*:

- Name it properly!

Example 22 Evaluate by "talking it out." (a) $\log_2 16$: "the _____" = _____

(b) $\log_8 \frac{1}{8}$ = _____ (c) $\log_{36} 6$ = _____ (d) $\log_{63} 1$ = _____ (e) $\log_5 5$ = _____

Two Identities

1. $\log_b 1 =$
2. $\log_b b = 1$

Inverse Identities: Since $f(x) = b^x$ and $g(x) = \log_b x$ are inverses, composing them *MUST* result in *x*:

$$f(g(x)) = f(\log_b x) = b^{\log_b x} = x \quad \text{and} \quad g(f(x)) = g(b^x) = \log_b b^x = x$$

Inverse Identities

$$b^{\log_b x} = x$$
$$\log_b b^x = x$$

Example 23 Simplify (a) $\log_3 3^8$ = _____ (b) $3\ln e^{\pi}$ = _____ (c) $8^{\log_8 5}$ = _____ (d) $10^{\log e}$ = _____

(e) $\log\sqrt{10}$ (f) $\ln(\frac{1}{e^2})$ (g) $\log_3(\frac{1}{27})$ (h) $\log_8\left(\frac{1}{2}\right)$

I) $\log_{81} \frac{1}{27} =$ **J)** $\log_{\frac{1}{16}} 32 =$ **K)** $\log_4 0 =$ **L)** $\log_{10} 1 =$

M) $\log_4 \frac{1}{8} =$ **N)** $\log_{27} \frac{1}{3} =$ **O)** $\log_9 3 =$ **P)** $\log_6 6^{3x} =$

Q) $\log_{36} \frac{1}{6} =$ **R)** $\log_{128} 2 =$ **S)** $\log_{\frac{1}{4}} 16 =$ **T)** $\log_z z^{2x} =$

U) $\ln e^{12} =$ **V)** $3^{\log_3 5} =$ **W)** $\ln 1 =$ **X)** $e^{\ln 4x} =$

The 3 Laws of Logarithms:

For any positive real numbers M and N and a real number p:

1. $\log_b MN = \log_b M + \log_b N$	(Product Rule)
2. $\log_b \frac{M}{N} = \log_b M - \log_b N$	(Quotient Rule)
3. $\log_b M^p = p\log_b M$	(Power Rule)

Change of Base Theorem:

$$\log_b M = \frac{\log_a M}{\log_a b}$$

Example 24A Use the stated Rule.

(a) *Product Rule:* $\log_2 3 + \log_2 1 =$

(b) *Quotient Rule:* $\log_2 \frac{3}{W} =$

(c) *Power Rule:* $\log_2 3^w =$

Example 24B Write as a base e log expression:

$\log_2 3 =$

Example 25 Write as a single logarithm with coefficient 1 and simplify as much as possible.

(a) $3\log_2 x + \log_2 10$

(b) $\frac{1}{2}\log 16x^2 - \log x + \log(x-1)$

Example 26 Separate $\log_2 \frac{8x^5}{(x-1)^3}$ into simpler logs.

Example 27

True or False?

(1) $\log\left(\frac{x}{y^3}\right) = \log x - 3\log y$

(2) $\log(a-b) = \log a - \log b$

(3) $\log x^k = k \cdot \log x$

(4) $(\log a)(\log b) = \log(a+b)$

(5) $\frac{\log a}{\log b} = \log(a-b)$

(6) $(\ln a)^k = k \cdot \ln a$

(7) $\log_a a^a = a$

(8) $-\ln\left(\frac{1}{x}\right) = \ln x$

(9) $\ln_{\sqrt{x}} x^k = 2k$

9.5-9.6 Exponential & Log Equations

1-1 Property for Exponents

If $b^M = b^N$, then $M = N$

1-1 Property for Logarithms

If $\log_b M = \log_b N$, then $M = N$

and

If $M = N$, then $\log_b M = \log_b N$

Exponent Rules:

$b^x b^y = b^{x+y}$

$(b^x)^y = b^{xy}$

Solving *basic* Exponential Equations

The object is to get the variable out of the exponent position.

1. Use the 1-1 Property for Exponents if both sides of the equation can be written in terms of the same base.
2, Convert to log form if there's only 1 exponential expression equaling a number and the base of the exponent is *e* or 10.
3. Use the 1-1 Property for Logs to take the "log of both sides." Then use the 3rd law of logarithms to bring the exponent down.

Example 28 Solve each equation. All answers containing logs must be in terms of base *e* or 10.

(a) $4^{x-3} = \frac{8}{2^{x-1}}$

(b) $e^{-x^2} = (e^x)^2 \cdot \frac{1}{e^3}$

(c) $5e^{3x+5} = 1$

(d) $3^{x+5} - 6 = 1$

(e) $5^{2x-3} = 3^{x-1}$

(f) $6 = \frac{18}{1+3e^{-0.2t}}$

Solving *basic* Logarithmic Equations: The object is to "free" the variable from the inside of the logarithm or from its base.
Use the law of logs to create one of 2 situations: 1. One log equals a number - Solve by converting to exponential form.
2. One log equals one log - Use the 1-1 property to *equate* the insides of the logarithms.

Example 29 Solve each equation. All answers containing logs must be in terms of base e or 10.

(a) $\frac{1}{2}\log_{16} x = \frac{1}{8}$

(b) $2\ln(x-1)+3=7$

(c) $2\log\sqrt{x+10} = 2-\log(x-5)$

(d) $\log(x-1)-2\log 3 = \log(6-x)$

***Other* Equations** If all terms have a common factor, or if it's a quadratic in terms of some exponential expression, then **FACTOR!**

(e) $2e^{2x}-7e^{x}+3=$

(f) $2e^{3x}(\ln x)^2-4e^{3x}\ln x=$

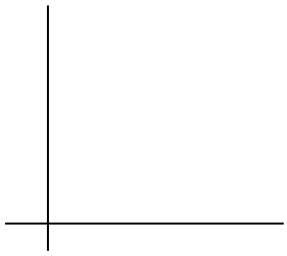
9.7 Exponential Applications

Exponential Growth and Decay: A population or quantity that experiences exponential growth (or decay) increases (or decreases) according to the model $y = y_0e^{kt}$ where y is the population or quantity at time t, y_0 is the initial size or amount of y, and k is the relative rate of growth or decay. We have growth when $k > 0$, and decay when $k < 0$.

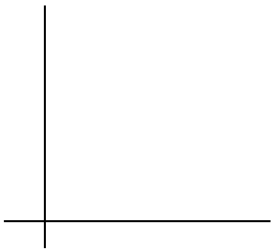
Radioactive Decay: The **half-life** of a radioactive substance is the time it takes for half the substance to decay.

Model: $y = y_0e^{kt}$

Exponential growth, $k > 0$



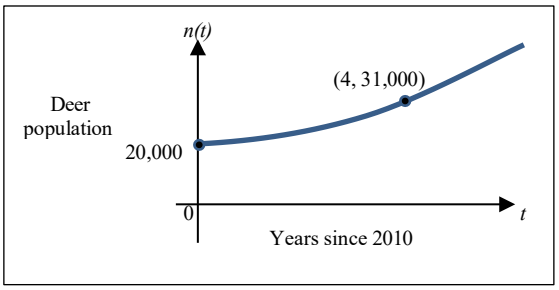
Exponential decay, $k < 0$



Example 30 The number of bacteria in a culture is modeled by the function $P(t) = 500\,e^{0.4t}$ where t is measured in hours.
(a) The initial number of bacteria is _____.
(b) The relative growth rate of this bacterium population (as a percentage) is _____.
(c) After how many hours will the population reach 10,000?

Example 30 The radioactive isotope gallium 67, used in the diagnosis of malignant tumors, has a biological half-life of 46.5 hours. Suppose a patient is injected with 2 mg of gallium 67.
(a) Write a function of the form $Q(t) = Q_0e^{-kt}$ to model the quantity of gallium 67 remaining after t hours.
(b) When will only 0.1 mg of gallium 67 remain?

Example 31 The graph shows the deer population in a Pennsylvania county between 2010 and 2014. Assume that the population grows exponentially. Find a function that models the deer population *t* years after 2010.



Logistic Growth

Example 32 The population of a certain fish in a stocked pond is given by $P(t) = \frac{8000}{1+3e^{-0.8t}}$ where *t* is time in years.

(a) The initial population is _____

(d) How long until the fish population reaches 4,000?

(b) What is the carrying capacity for the fish? _____

(c) Graph $P(t)$.

Compound Interest $A = P\left(1 + \frac{r}{n}\right)^{nt}$ where A is the amount at the end of *t* years when a principle P is invested at an annual rate *r* compounded *n* times a year.

Continuous Compound Interest $A = Pe^{rt}$ where A is the amount at the end of *t* years when a principle P is invested at an annual rate *r* compounded continuously.

Example 33 If compounded quarterly, at what rate would $10,000 have to be invested in order for its value to reach $14,000 in 10 years?

Example 34 Suppose that $1100 is initially invested in an account at a fixed interest rate, compounded *continuously*. Suppose also that, after five years, the amount of money in the account is $1302. Find the interest rate per year. Write your answer as a percentage. Do not round any intermediate computations, and round your percentage to the nearest hundredth.